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Evaluating instances

Introduction

For the last four chapters, we have been studying concordances in one form or another. Each instance has been taken to be as important as any other, and has had to be accounted for. This is a valuable discipline, but only the very first step towards the automation of text study. In this chapter and the next, we begin to evaluate concordances and devise new kinds of information about language.

The starting point of this chapter is probably unexpected—it is that most actual examples are unrepresentative of the pattern of the word or phrase for which they are chosen.

Such is the intricate nature of the ties between one segment of text and the surrounding text, and the relation between the text and the world and the intended outcomes of the communication, that the act of plucking a few words from any text is not likely to provide a freestanding instance of its constituent words, each acting typically.

The vast majority can be safely discarded when their statistical contribution to the concordance as a whole has been recorded. We need a lot of text so that there will always be a sufficient residue of useful examples, and also to provide criteria for discarding the others in the first place.

Throw away your evidence

The policy of discarding examples, and particularly examples which do not fit a description, is likely to have to struggle for popularity in linguistics. The Cult of the Counter-example is still very strong, in myth if not always in observance, and it is important for students of text to define a careful position in this regard, which will be quite different from that of students of sentences. For example, the computer corpora of the early sixties, Brown (Kučera and Francis 1967) and LOB (Hofland and Johansson 1982), represent a transitional stage; they

were most carefully constructed in an attempt to be representative, and each instance is cherished. The corpora are just small enough (one million words) for this to be possible for determined scholars.

The received wisdom of corpus linguistics is that fairly small corpora, of one million words or even fewer, are adequate for grammatical purposes, since the frequency of occurrence of so-called grammatical or function words is quite high. In the LOB corpus, for example (one million words of English printed in the UK in the year 1961), the commonest 100 words are almost all grammatical, and range in frequency from *the* at 68,315 to *people* at 953 (the 'lexical' words in that range are *said* 2,074, *time* 1,654, *man* 1,072, *years* 1,067). There are a few stragglers, like *shall* 348, *itself* 272, *nor* 200, *else* 169, down to *whatsoever* 7, and *whichsoever* 1; but there are numerous instances of grammatical words, sufficient to enable conventional grammatical statements to be made.

However, the availability nowadays of much larger corpora makes it possible to evaluate conventional grammatical statements. Presumably, the shorter original corpora did little more than confirm the generally agreed positions on English grammar. The new evidence suggests that grammatical generalizations do not rest on a rigid foundation, but are the accumulation of the patterns of hundreds of individual words and phrases. The language looks rather different when you look at a lot of it at once.

Such evidence has not been available before. Linguists have had to rely on their intuitions, their limited capacity for thorough textual analysis, and whatever has caught their eye or ear as they have encountered large extents of language behaviour, in their daily lives or in their professional work.

The great dictionaries of English used human beings to evaluate their examples, and there is as yet no substitute. This method is likely to highlight the unusual in English and perhaps miss some of the regular, humdrum patterns.

In grammars, the tradition of citing examples has faded away since *A Modern English Grammar* (Jespersen 1909; 1949). Even *A Comprehensive Grammar of the English Language* (CGEL) (Quirk *et al.* 1985) relies heavily on invented examples.

The CGEL had the corpus of the Survey of English Usage available to it: this is a corpus approaching one million words, spanning twenty-five years, and including a substantial proportion of spoken English.

Occasional reference is given in the grammar to Survey material, but no attempt is made to confront and account for the evidence. Hardly any of the examples are citations, though citations must have been readily available. One is forced to conclude that the authors were following a methodology which gave low priority to one of the concerns of this book, which is to press for the use of actual language data as a basis for all descriptive statements.

A valid generalization about data must relate to the data in a systematic way; each relevant instance must either support the generalization or exhibit features which make the generalization subordinate to some other descriptive statement. Hence, it is important to fix on a particular body of data (which is best chosen on non-linguistic principles), and then engage with every instance. If that procedure is adopted in language work, it soon becomes necessary to acquire very large quantities of data, or else generalizations cannot be made. Language is very complex, and people use it for their own ends, without normally being conscious of the relation between their verbal behaviour and the way that behaviour is characterized. They are creative, or expedient, or casual, or confused; or they have unusual matters to put into usual words, so they have to combine them in unusual ways.

It is, therefore, necessary to have access to a large corpus because the normal use of language is highly specific, and good representative examples are hard to find. This is as true of grammar as of lexis, because grammar is not made of just the patterns of the common grammatical words, but relies on the whole vocabulary of the language.

One further factor makes it essential to collect a large number of instances. Many words have more than one meaning, sense, or usage, and these occur in very uneven distribution. As far as I know, no systematic research has yet been done in this area, so the following remarks are speculations based on observation and occasional probings.

Frequent words have, in general, a more complex set of senses than infrequent words. If we divide and number senses in the conventional dictionary manner, we may discover a statistical relationship between the number of occurrences of a word and the number of different senses it realizes. Hence, the accumulation of instances of a frequent word is not just more of the same, but ever more clear evidence of complexity.

In addition to this, we must allow that, just as some words are much more common than others, some senses of one word are much more

common than other senses of the same word—many times more common. So if we need, say, fifty occurrences of a sense of a word in order to describe it thoroughly, then the corpus has to be large enough to yield fifty instances of the least common sense. In practice, we find that the decision about the 'least common sense' is an arbitrary one; no matter what the size, there are always loose ends, unstable senses, occasional odd examples, etc. But wherever this limit is fixed, we shall observe huge discrepancies in the frequency of the recognized senses, and this will produce a heavy demand for very long texts.

Text and language

The distinction has often been made between text and language on a dimension of abstraction. Language is an abstract system; it is realized in text, which is a collection of instances. This is clearly an inadequate point of view, because we do not end up with anything like text by 'generating' word strings from grammars. In particular, there is hardly any allowance for the combinatorial meanings in text. If text (including, and in particular, spoken text) is not a strict realization of meaningful abstract decisions, then either it is subject to random distortion, or it is in part the result of decisions which are not recorded in the abstract system, but which take precedence over those which are. Many of these are components of the rather mystical notion of 'coherence' that is beyond the generative competence of grammars. Random factors will certainly not explain how coherence arises; so we are forced to conclude that the realization route is not through conventional grammars, but through some kind of functional analysis.

Actual text will always be deviant with respect to structural rules of the conventional kind. Some of the factors that lead to deviance have already been mentioned—creativity, unselfconsciousness, expediency, inattention, confusion, and the need to express the unusual. Another major factor is shared knowledge among communicators, which leads to the actual occurrence of many utterances which are proscribed by rule (for example, an obligatory transitive verb occurring without an object where the real-world thing that could give rise to an expressed object is obvious).

The grammarian's dilemma is this: does he or she study actual instances, knowing that most of them are untypical, or does he or she ignore actual instances and study a set of instances which have not

occurred, including a high proportion that could never occur? The former choice characterized the field linguists in the first half of this century, and the latter choice has been evident in the linguistics of the last thirty years.

The new option opened up by the computer is to evaluate actual instances and select the most typical. A complete set of typical instances should exemplify the dominant structural patterns of the language without recourse to abstraction, or indeed to generalization. The mass of instances each contain just a small element of typicality, but a few contain several typical features. In such circumstances, although it may sound paradoxical, examples which are typical are rather uncommon, and have to be found by statistical methods.

It is, therefore, unnecessary to make a sharp distinction between abstract and actual language structure—the sort of distinction embodied in Saussure's *langue* and *parole* or Chomsky's *competence* and *performance*. The existence of these dichotomies is to allow us to abstract from the chaos of life a system of meaningful choices and to insulate the abstract system. I have already conceded that some proportion of the complexity of text may be attributable to accidental or random factors, but that is far from sufficient explanation. It may indeed have obscured what actually goes on. In fact, the main simplification that is introduced by conventional grammar has nothing to do with the purity of abstraction as against the chaos of life. It is merely the decoupling of lexis and syntax.

In the explicit theoretical statement of linguistics, grammatical and lexical patterns vary independently of each other. In most grammars, it is an assumption that is obviously taken for granted. For example, it is rare for a grammar to note that a certain structure is only appropriate for a particular sense of a word. The same goes for morphology. In contrast, grammars attribute independent meaning to syntactic arrangements.

Equally, it is rare for a dictionary to note the common syntactic patterns of a word in a particular sense. Pedagogical dictionaries are increasingly seeing this as essential information for learners, but it is added in the form of afterthoughts such as usage notes. The implicit stance of a conventional dictionary entry is that most of the words in daily use have several meanings, and any occurrence of the word could signal any one of the meanings. If this were actually the case, communication would be virtually impossible.

The decoupling of lexis and syntax leads to the creation of a rubbish dump that is called 'idiom', 'phraseology', 'collocation', and the like. If two systems are held to vary independently of each other, then any instances of one constraining the other will be consigned to a limbo for odd features, occasional observations, usage notes, etc. But if evidence accumulates to suggest that a substantial proportion of the language description is of this mixed nature, then the original decoupling must be called into question. The evidence now becoming available casts grave doubts on the wisdom of postulating separate domains of lexis and syntax.

In modern lexical research, it is part of the long-term task to specify accurately the established phrases of a language. A phrase can be defined for the moment as a co-occurrence of words which creates a sense that is not the simple combination of the sense of each of the words. One is first struck by the fixity and regularity of phrases, then by their flexibility and variability, then by the characteristically creative extensions and adaptations which occur, sometimes more often than the 'ordinary' form.

In this work, it is much more fruitful to start by supposing that lexical and syntactic choices correlate, than that they vary independently of each other.

Meaning and structure

For the remainder of this chapter (as in Chapter 4), I should like to widen the domain of syntax to include lexical structure as well, and call the broader domain *structure*. In the spirit of the preceding argument, I shall define structure as any privileges of occurrence of morphemes; we do not in the first analysis have to decide whether these are lexical or syntactic—or as so often—a bit of both.

Is it then best to hypothesize that sense and structure are inseparable? Unfortunately not. If that were so, ambiguity would be impossible. More than one sense can be realized by the same structure, and, in the simplest case, by the same word.

We must, then, consider whether ambiguity is incidental or systematic. If it is much more than incidental and occasional, then it will constitute strong evidence for the independence of lexis and syntax. However, although ambiguity causes great headaches in automatic parsing, if we look at the way people actually operate with language we see it as a sporadic and almost accidental coincidence of realization, rarely constituting a communicative problem. Much more of it

would threaten the basic notion of realization in language—that structure realizes sense and therefore normally differentiates one sense from another.

If sense and structure are not independent of each other and not inseparable, then they must be associated. Here we can frame a hypothesis that can act as a substitute for the langue/parole distinction. We can postulate that the underlying unit of composition is an integrated sense-structure complex, but that the exigencies of text frequently obscure this. This position offers a sharp contrast to the atomistic model featured by most grammars, and the argument is developed in the next chapter.

Our descriptive task then becomes the identification of the regular and typical associations, leading to the identification of one or more 'citation forms' for each distinct sense. The distinguishing features of the citation forms could then be stated, and explanations could be offered for the occurrence of non-citation forms. A citation form would involve a modest step in abstraction. It is also likely that many citation forms contain some systematic variables, such as pronoun selections, which leaves a modicum of independence to the grammar.

Procedure

How, then, do we find the citation forms, especially since we believe text to be largely composed of non-citation forms? I propose to outline a method for tackling one area of structure, in this case collocation, which gives promise of valuable results. The same principle can be applied to other structural features.

The procedure begins with a machine-generated concordance to a large corpus, as we have used in previous studies in this book. The usual kind of concordance is adequate, where all the occurrences of a word-form are retrieved, each in the middle of a line of text. A line of text may contain as many as eight or nine words on either side of the central word, or *node*, and we do not expect to need more than four or five on either side.

A concordance has many of the properties of a natural text, and it is reasonable for the purposes of statistical analysis to treat each cited line as if it were a sentence, and so to examine the vocabulary of the concordance. In order to do this, a list is compiled in frequency order, of all the word-forms in the concordance. These are called the *collocates* of the node. This raw list is then processed as follows:

- a. The lines are trimmed so that only those words that are reasonably likely to be attracted by the node are left in. It was strongly suggested in Sinclair, Jones, and Daley (1970) that beyond four words from the node there were no statistical indications of the attractive power of the node. At present we are experimenting with environments of between one and five words on either side of the node, both balanced and unbalanced, to see if there is an optimum setting.
- b. There is no point in considering very infrequent collocates, and there is usually a long tail to the frequency lists. A suitable cut-off point—for example, less than ten per cent of the frequency of the node—should be determined.
- c. Each of the remaining collocates is given a weighting by relating its frequency in the concordance to its overall frequency in the full corpus. So a common word gets a low rating, and a word which makes a distinctive collocation with the node will score high.
- d. Each line of the concordance is now examined for the typicality of its collocates, by adding up the weightings of each collocate in its environment. The concordance is now re-sorted into an order of typicality, and the most typical instances should come to the top.

From this point onwards, an automatic procedure is not yet fully established, and the study continues largely on a subjective basis. First, any obvious phrases are identified and removed. Next, there is a search for the clustering of collocates and their mutual attraction and repulsion, for example, pairs and groups of collocates which frequently occur in the same line, and pairs which never do. Then the other aspects of structural patterning are brought in—the occurrence of the very frequent words, syntactic structures, ordering of items in the line, and so on.

If it is suspected that there are two or more principal senses of a word, an attempt is made to isolate a sense, using explicit criteria. When a sense is fully described, all the lines that exemplify it are then removed, and the new, shorter concordance is reprocessed from the beginning.

Gradually, this procedure should identify the distinct senses of a word. Each cycle will, however, reduce the size of the remaining concordance substantially, and the overall size of the corpus will quickly become a limiting factor.

Findings

This technique, in a provisional form, was recently applied to the concordance of the word *second*. The word was chosen as being fairly frequent (over 1,200 occurrences in 7.3 million), and as having two rather distinctive major senses. It was found that the first pass identified *the Second World War* as a phrase which had 14 occurrences in the 50 most typical. The next pass, omitting the 14 phrases, identified a major sense which was strongly associated with preceding *the*, occasionally *his* or *her*, and with words like *first*, *third*, *time*, *year*, *act*, *child*, and *wife* in the environment. The next pass identified a sense which was strongly associated with preceding *per*, and before that a word like *cycles*, *radians*. A number of similar instances had *a* instead of *per* but *a* is also occasionally used in the other main sense.

There was little else except a hint of possible phrases *second hand* and *second class*.

The two main meanings of *second*, then, are associated one with definiteness and the other with indefiniteness. This is at least as important as the observation that one is a modifier and one a noun.

A closer look at the full concordance confirms these findings. There is, however, a third fairly prominent use of *second* which does not emerge in the collocational analysis. This requires neither a definite nor an indefinite determiner, and the word functions as a discourse organizer. It is quite often preceded by *and*. It is not surprising that this use does not attract strong lexical collocations, because it occurs according to the exigencies of the discourse and should be largely independent of such things as content, topic, message.

These findings are crude, preliminary, and partial. No doubt a study of *secondly* would identify the third sense of *second* as a discourse organizer to be absorbed into the lemma *second(ly)*. The study of *seconds* might add new features and new uses, and so on. In due course, we shall see. The technique has at least managed to isolate the most basic contrast of meaning. In another trial at an international conference in 1988, the prototype system successfully distinguished among *sole* = only, *sole* = fish, and *sole* = bottom of shoes or feet.

Conclusion

The conclusion that can be drawn from this and other examples is that it is folly to decouple lexis and syntax, or either of those and semantics. The realization of meaning is much more explicit than is suggested by abstract grammars. The model of a highly generalized formal syntax, with slots into which fall neat lists of words, is suitable only in rare uses and specialized texts. By far the majority of text is made of the occurrence of common words in common patterns, or in slight variants of those common patterns. Most everyday words do not have an independent meaning, or meanings, but are components of a rich repertoire of multi-word patterns that make up text. This is totally obscured by the procedures of conventional grammar.

The next chapter takes up this argument in detail. The notion of citation forms is developed in a separate publication (Sinclair 1984).

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Collocation

Introduction

This chapter concludes the description of word co-occurrence as we currently conceive it. The next stage is to write a dictionary of collocations, and the project is in hand (Sinclair *et al.* forthcoming). The argument brings together a number of themes that have been developing throughout the book, in particular, the notions of dependent and independent meaning, and the relation of texts to grammar.

Two models of interpretation

It is contended here that in order to explain the way in which meaning arises from language text, we have to advance two different principles of interpretation. One is not enough. No single principle has been advanced which accounts for the evidence in a satisfactory way. The two principles are.

The open-choice principle

This is a way of seeing language text as the result of a very large number of complex choices. At each point where a unit is completed (a word or a phrase or a clause), a large range of choice opens up and the only restraint is grammaticalness.

This is probably the normal way of seeing and describing language. It is often called a 'slot-and-filler' model, envisaging texts as a series of slots which have to be filled from a lexicon which satisfies local restraints. At each slot, virtually any word can occur. Since language is believed to operate simultaneously on several levels, there is a very complex pattern of choices in progress at any moment, but the underlying principle is simple enough.

Any segmental approach to description which deals with progressive choices is of this type. Any tree structure shows it clearly: the nodes on

the tree are the choice points. Virtually all grammars are constructed on the open-choice principle.

The idiom principle

It is clear that words do not occur at random in a text, and that the open-choice principle does not provide for substantial enough restraints on consecutive choices. We would not produce normal text simply by operating the open-choice principle.

To some extent, the nature of the world around us is reflected in the organization of language and contributes to the unrandomness. Things which occur physically together have a stronger chance of being mentioned together; also concepts in the same philosophical area, and the results of exercising a number of organizing features such as contrasts or series. But even allowing for these, there are many ways of saying things, many choices within language that have little or nothing to do with the world outside.

There are sets of linguistic choices which come under the heading of register, and which can be seen as large-scale conditioning choices. Once a register choice is made, and these are normally social choices, then all the slot-by-slot choices are massively reduced in scope or even, in some cases, pre-empted.

Allowing for register as well, there is still far too much opportunity for choice in the model, and the principle of idiom is put forward to account for the restraints that are not captured by the open-choice model.

The principle of idiom is that a language user has available to him or her a large number of semi-preconstructed phrases that constitute single choices, even though they might appear to be analysable into segments. To some extent, this may reflect the recurrence of similar situations in human affairs; it may illustrate a natural tendency to economy of effort; or it may be motivated in part by the exigencies of real-time conversation. However it arises, it has been relegated to an inferior position in most current linguistics, because it does not fit the open-choice model.

At its simplest, the principle of idiom can be seen in the apparently simultaneous choice of two words, for example, *of course*. This phrase operates effectively as a single word, and the word space, which is structurally bogus, may disappear in time, as we see in *maybe*, *anyway*, and *another*.

Where there is no variation in the phrase, we are dealing with a fairly trivial mismatch between the writing system and the grammar. The *of*

in *of course* is not the preposition *of* that is found in grammar books. The preposition *of* is normally found after the noun head of a nominal group, or in a quantifier like *a pint of ...*. In an open-choice model, *of* can be followed by any nominal group (see Chapter 6 for details). Similarly, *course* is not the countable noun that dictionaries mention; its meaning is not a property of the word, but of the phrase. If it were a countable noun in the singular it would have to be preceded by a determiner to be grammatical, so it clearly is not.

It would be reasonable to add phrases like *of course* to the list of compounds, like *cupboard*, whose elements have lost their semantic identity, and make allowance for the intrusive word space. The same treatment could be given to hundreds of similar phrases—any occasion where one decision leads to more than one word in text. Idioms, proverbs, clichés, technical terms, jargon expressions, phrasal verbs, and the like could all be covered by a fairly simple statement.

However, the principle of idiom is far more pervasive and elusive than we have allowed so far. It has been noted by many writers on language, but its importance has been largely neglected. Some features of the idiom principle follow:

- a. Many phrases have an indeterminate extent. As an example, consider *set eyes on*. This seems to attract a pronoun subject, and either *never* or a temporal conjunction like *the moment*, *the first time*, and the word *has* as an auxiliary to *set*. How much of this is integral to the phrase, and how much is in the nature of collocational attraction?
- b. Many phrases allow internal lexical variation. For example, there seems to be little to choose between *in some cases* and *in some instances*; or between *set x on fire* and *set fire to x*.
- c. Many phrases allow internal lexical syntactic variation. Consider the phrase *it's not in his nature to ...*. The word *it* is part of the phrase, and so is the verb *is*—though this verb can vary to *was* and perhaps can include modals. *Not* can be replaced by any 'broad' negative, including *hardly*, *scarcely*, etc. *In* is fixed, but *his* can be replaced by any possessive pronoun and perhaps by some names with 's. *Nature* is fixed.
- d. Many phrases allow some variation in word order. Continuing the last example, we can postulate *to recriminate is not in his nature*, or *it is not in the nature of an academic to ...*

- e. Many uses of words and phrases attract other words in strong collocation; for example, *hard work*, *hard luck*, *hard facts*, *hard evidence*.
- f. Many uses of words and phrases show a tendency to co-occur with certain grammatical choices. For example, it was pointed out in Chapter 5 that the phrasal verb *set about*, in its meaning of something like 'inaugurate', is closely associated with a following verb in the *-ing* form, for example, *set about leaving* ... What is more, the second verb is usually transitive, for example, *set about testing it*. Very often, *set* will be found in co-occurrence patterns.
- g. Many uses of words and phrases show a tendency to occur in a certain semantic environment. For example, the verb *happen* is associated with unpleasant things—accidents and the like.

The overwhelming nature of this evidence leads us to elevate the principle of idiom from being a rather minor feature, compared with grammar, to being at least as important as grammar in the explanation of how meaning arises in text. Support comes unexpectedly from a different quarter.

Evidence from long texts

In the current lexical analysis of long texts, a number of problems have arisen, not all of which were anticipated:

- 1 The 'meanings' of very frequent, so-called grammatical words are a headache in any lexicography, but the problem they typify fits in with some of the newer difficulties.
- 2 Some 'meanings' of frequent words seem to have very little meaning at all, for example, *take*, in *take a look at this*; *make* in *make up your mind*.
- 3 The commonest meanings of the commonest words are not the meanings supplied by introspection; for example, the meaning of *back* as 'the posterior part of the human body, extending from the neck to the pelvis' (*Collins English Dictionary* (CED) 2nd edition 1986 sense 1) is not a very common meaning. Not until sense 47, the second adverbial sense, do we come to 'in, to or towards the original starting point, place or condition', which is closer to the commonest usage in our evidence.

I think most speakers of English would agree with the CED's ordering of senses, whatever the evidence from frequency. What is disquieting is the apparent lack of good reason for the enormous discrepancy

between the sense to which our intuitions give priority, and the most frequent one.

- 4 The commonest meanings of many less common words are not those supplied by introspection. Sense 1 offered in the CED for *pursue* is 'to follow (a fugitive etc.) in order to capture or overtake', yet by far the commonest meaning is sense 5, 'to apply oneself to (one's studies, hobbies, interests etc.)'.

From this evidence, we can put forward some tentative generalizations:

- 1 There is a broad general tendency for frequent words, or frequent senses of words, to have less of a clear and independent meaning than less frequent words or senses. These meanings of frequent words are difficult to identify and explain; and, with the very frequent words, we are reduced to talking about uses rather than meanings. The tendency can be seen as a progressive delexicalization, or reduction of the distinctive contribution made by that word to the meaning.
- 2 This dependency of meaning correlates with the operation of the idiom principle to make fewer and larger choices. The evidence of collocation supports the point. If the words collocate significantly, then to the extent of that significance, their presence is the result of a single choice.
- 3 The 'core' meaning of a word—the one that first comes to mind for most people—will not normally be a delexical one. A likely hypothesis is that the 'core' meaning is the most frequent independent sense. This hypothesis would have to be extensively tested, but if it proved to hold good then it would help to explain the discrepancy referred to above between the most frequent sense and what intuition suggests is the most important or central one.
- 4 Most normal text is made up of the occurrence of frequent words, and the frequent senses of less frequent words. Hence, normal text is largely delexicalized, and appears to be formed by exercise of the idiom principle, with occasional switching to the open-choice principle.
- 5 Just as it is misleading and unrevealing to subject *of course* to grammatical analysis, it is also unhelpful to attempt to analyse grammatically any portion of text which appears to be constructed on the idiom principle.

The last point contains an implication that a description must indicate how users know which way to interpret each portion of an utterance. The boundaries between stretches constructed on different principles will not normally be clear-cut, and not all stretches carry as much evidence as *of course* does to suggest that it is not constructed by the normal rules of grammar.

It should be recognized that the two models of language that are in use are incompatible with each other. There is no shading of one into another; the switch from one model to the other will be sharp. The models are diametrically opposed.

The last two points taken together suggest one reason why language text is often indeterminate in its interpretation and hence very flexible in use. If the 'switch points' between two modes of interpretation are not always explicitly signalled, and the two modes offer sharply contrasting ways of interpreting the data, then it is quite likely that an utterance will not be interpreted in exactly the same way in which it was constructed. Also, two listeners, or two readers, will not interpret in precisely the same way.

For normal texts, we can put forward the proposal that the first mode to be applied is the idiom principle, since most of the text will be interpretable by this principle. Whenever there is good reason, the interpretive process switches to the open-choice principle, and quickly back again. Lexical choices which are unexpected in their environment will presumably occasion a switch; choices which, if grammatically interpreted, would be unusual are an affirmation of the operation of the idiom principle.

Some texts may be composed in a tradition which makes greater than normal use of the open-choice principle; legal statements, for example. Some poems may contrast the two principles of interpretation. But these are specialized genres that require additional practice in understanding.

It thus appears that a model of language which divides grammar and lexis, and which uses the grammar to provide a string of lexical choice points, is a secondary model. It cannot be relinquished, because a text still has many switch points where the open-choice model will come into play. It has an abstract relevance, in the sense that much of the text shows a potential for being analysed as the result of open choices, but the other principle, the idiom principle, dominates. The open-choice analysis could be imagined as an analytical process which goes on in principle all the time, but whose results are only intermittently called for.

This view of how the two principles are deployed in interpretation can be used to make predictions about the way people behave, and the accuracy of the predictions can be used as a measure of the accuracy of the model. Areas of relevant study include: the transitional probabilities of words; the prevalent notion of *chunking* (see Chapter 9); the occurrence of hesitations, etc., and the placement of boundaries; and the behaviour of subjects trying to guess the next word in a mystery text.

Collocation

The above is the framework within which I would like to consider the role of collocation. Collocation, as has been mentioned, illustrates the idiom principle. On some occasions, words appear to be chosen in pairs or groups and these are not necessarily adjacent.

One aspect of collocation has been of enduring interest. When two words of different frequencies collocate significantly, the collocation has a different value in the description of each of the two words. If word *a* is twice as frequent as word *b*, then each time they occur together is twice as important for *b* than it is for *a*. This is because that particular event accounts for twice the proportion of the occurrence of *b* than of *a*.

So when all the occurrences of *a* with *b* are counted up and evaluated, one figure is recorded in the profile of *a*, and another figure double the size, is recorded in the profile of *b*.

By entering the same set of events twice, once as the collocation of *a* with *b* and again as the collocation of *b* with *a*, one incurs the strictures of Benson, Brainerd, and Greaves (1985) who say 'there are two problems here: double counting of nodes and double counting of collocates. The parts now add up to considerably more than the whole, which makes computation under any statistical model inaccurate'. In practice, the possibility of double entry allows us to highlight two different aspects of collocation.

I would like to consider separately the two types of collocation instanced above, using the term *node* for the word that is being studied, and the term *collocate* for any word that occurs in the specified environment of a node. Each successive word in a text is thus both node and collocate, though never at the same time.

When *a* is node and *b* is collocate, I shall call this *downward collocation*—collocation of *a* with a less frequent word (*b*). When *b* is

node and *a* is collocate, I shall call this *upward collocation*. The whole of a given word list may be treated in this way.

There appears to be a systematic difference between upward and downward collocation. Upward collocation, of course, is the weaker pattern in statistical terms, and the words tend to be elements of grammatical frames, or superordinates. Downward collocation by contrast gives us a semantic analysis of a word.

Collocation of *back*

Let us illustrate collocational patterns, in a provisional way, with the word *back*. I shall make no attempt to differentiate separate senses, but will put the collocates into *ad hoc* groups.

No standard of statistical significance is claimed at present, because many typical collocations are of such low frequency compared with the overall length of a text. Because of the low frequency of the vast majority of words, almost any repeated collocation is a most unlikely event, but because the set of texts is so large, unlikely events of this kind may still be the result of chance factors.

However, no speaker of English would doubt the importance of these patterns. One recognizes them immediately, because they are features of the organization of texts; often subliminal, they cannot be reliably retrieved by introspection.

In distinguishing upward and downward collocation I have made a buffer area of (plus or minus) 15 per cent of the frequency of the node word. For example, let us take a word occurring 1,000 times; when it is examined as a node, collocates are grouped into:

- a. upward collocates—those whose own occurrence is over 115 per cent of the node frequency (that is, 1,150);
- b. neutral collocates—between 85 per cent and 115 per cent of the node frequency (in this instance, 850 and 1,150), this is the buffer area;
- c. downward collocates—less than 80 per cent (in this instance, 850).

Neutral collocates are added on an *ad hoc* basis to upward or downward groups, and are given round brackets. Since this has to be a summary account of a very large set of data, I have removed some items which seem to be of little general significance. These include personal names, contracted forms like *I'll*, and word-forms whose

co-occurrence with *back* is infrequent and carries no conviction of any general significance. Of the last category, the form *anger* only occurs in the title of the play *Look Back in Anger*.

The nouns and verbs listed below as collocating with *back* are representative only. Given the uncertainty at the limits of statistical significance, it could be more misleading to include doubtful contenders. Thus, while *get*, *go*, and *bring* are unlikely to be challenged, *beach*, *box*, and *bus* are much less convincing when the actual instances are examined.

The qualification for an instance being scrutinized is co-occurrence within four words of *back*, on either side, this being the cut-off point established some years ago (Jones and Sinclair 1974). No account is taken of syntax, punctuation, change of speaker, or anything other than the word-forms themselves.

No doubt the studies which succeed this one will sharpen up the picture considerably. For example, the evidence of *back* suggests that few intuitively interesting collocations cross a punctuation mark. But it would be unwise to generalize from the pattern of one word, particularly such an unusual one as *back*. Now that tagged and parsed texts are becoming available, the co-patterning of lexical and grammatical choices is open to research. But it is still important to draw attention to the strength of patterning which emerges from the rawest of unprocessed data.

In pushing forward into new kinds of observation of language, the computer is simultaneously pulling us back to some very basic facts that are often ignored in linguistics. The set of four choices, a,b,c,k, from the alphabet, arranged in the sequence b,a,c,k with nothing in between them, that is, *back*, is an important linguistic event in its own right, long before it is ascribed a word-class or a meaning. It is difficult for users of English to notice this, but it is the computer's starting point.

Analysis of the collocational pattern of *back*

Upward collocates: *back*

Prepositions/adverbs/conjunctions: *at*, (*down*), *from*, *into*, *now*, *on*, *then*, *to*, *up*, *when*

Pronouns: *her*, *him*, *me*, *she*, *them*, *we*

Possessive pronouns: *her*, *his*, *my*, (*your*)

Verb: *get*, (*go*), *got*

The meaning of *back* as 'return' attracts expressions of time and place; *after* and *where* are also prominent. The presence of four subject pronouns may have a more general explanation than anything to do with *back*, but the absence of *you* and *I* from the list may be worth pursuing. Possessive pronouns suggest the anatomical sense of *back* and would explain why *they* and *their* do not figure prominently. The two verbs *get* and *go* are superordinates of a large number of verbs of motion, many of which will be found in the downward collocates.

I have selected a few examples of these words to show the way in which the basic syntax of *back* is established. The sets of examples follow the four categories mentioned above:

It really was like being back *at* school
 He drive back *down* to the terrace
 When our parents came back *from* Paris
 I followed him back *into* the wood
 A hefty slap *on* the back
 He turned back to the bookshelf
 When can I have *him* back home, doctor?
 She went back to her typing
 It would be nice to have *them* back
 We went back to the bungalow
 She has gone back to *her* parents
 He want back into *his* office
 I ran back to *my* cabin
 Go back to *your* dormitory at once
 Now I must *get* back to work
 They go back to the same nest

Downward collocates: back

Verbs: *arrive, bring, etc., climbed, come, etc., cut, etc., dates, etc., drew, etc., drove, etc., fall, etc., flew, flung, handed, hold, etc., jerked, lay, etc., leaned, etc., looked, looking, etc., pay, pulled, etc., pushed, etc., put, ran, rocking, rolled, rush, sank, sat, etc., sent, etc., shouted, snapped, stared, stepped, steps, etc., stood, threw, traced, turned, etc., walked, etc., waved.*

Prepositions: *along, behind, onto, past, toward, towards*

Adverbs: *again, forth, further, slowly, straight*

Adjective: *normal*

Nouns: *camp, flat, garden, home, hotel, office, road, streets, village, yard*
bed, chair, couch, door, sofa, wall, window, feet, forehead, hair, hand, head, neck, shoulder, car, seat
mind, sleep,
kitchen, living room, porch, room.

The word-class groupings above are based on frequency with *back*; many words actually occur in more than one word-class. Verbs are given in their most frequent form. Note the preponderance of past tense verbs, reflecting the temporal meaning of *back*.

The prepositions and adverbs suggest some typical phrases with *back*, and the nouns are largely those of direction, physical space, and human anatomy. A few typical examples follow:

Verbs: You *arrive* back on the Thursday
 May *bring* it back into fashion
 We *climbed* back up on the stepladder
 They had *come* back to England
 She never *cut* back on flowers
 It possibly *dates* back to the war
 The bearer *drew* back in fear
 We *drove* back to Cambridge
 You can *fall* back on something definite
 I *flew* back home in a light aircraft
 He *flung* back the drapes joyously
 Don't try to *hold* her back
 She *lay* back in the darkness
 He *leaned* back in his chair
 He *looked* back at her, and their eyes met
 Pay me back for all you took from me
Pulled back the bedclothes and climbed into bed
 I *pushed* back my chair and made to rise
 Shall I *put* it back in the box for you
 I *rolled* back onto the grass
 She *sat* back and crossed her legs
 Edward was *sent* back to school
 He *shouted* back
 The girl *stared* back
 They *started* walking back to Fifth Avenue

He *stepped* back and said ...
 He then *stood* back for a minute
 The woman *threw* her head back
 These could be *traced* back to the early sixties
 He *turned* back to the bookshelf
 She *walked* back to the bus stop
 We *waved* back like anything

Prepositions: Hands held *behind* his back
 Walked back *toward* the house

Adverbs: Later we came back *again*
 Rock us gently back and *forth*
 If you look *further* back in my files
 The *straight* back to his cabin
 He went *slowly* back to his book

Adjective: Things would soon get back to *normal*

Nouns: I crawled back to *camp*
 I'll drive you back to your *flat*
 Not a bit like his back *garden*
 He turned and went back *home*
 We had to go back to the *hotel*
 You've just got back from the *office*
 Set back from the *road*
 The back *streets* of Glasgow
 All the way back to the *village*
 On his *way* back to the apartment
 Without even a back *yard*
 Go back to *bed*
 He leaned back in his *chair*
 Stepping outside the back *door*
 A man standing by the back *wall*
 Tom went back to the *window*
 Britain would be back on its *feet*
 He brushed back his *hair*
 With the back of his *hand*
 She put her *head* back against the seat
 The hairs on the back of my *neck*
 He gestured back over his *shoulder*
 They got back into the *car*

There was some beer on the back *seat*
 In the back of his *mind*
 Then we go back to *sleep* again
 You must come back to the *kitchen*
 She went back into the *living room*
 Beside me here on the back *porch*
 He came back into the *room*

Conclusion

All the evidence points to an underlying rigidity of phraseology, despite a rich superficial variation. Hardly any collocates occur more than once in more than two patterns. The phraseology is frequently discriminatory in terms of sense; for example, there are almost as many instances of *flat on her back* as *back to her flat*. Some, like *arrive*, seem characteristic of the spoken language, some, like *hotel*, show the wisdom of allowing a nine-word span for collocation.

Early predictions of lexical structure were suitably cautious; there was no reason to believe that the patterns of lexis should map on to semantic structures. For one thing, lexis was syntagmatic and semantics was paradigmatic; for another, lexis was limited to evidence of physical co-occurrence, whereas semantics was intuitive and associative.

The early results given here are characteristic of present evidence; there is a great deal of overlap with semantics, and very little reason to posit an independent semantics for the purpose of text description.

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The rationale is set out in the Introduction. For a word in the Cobuild dictionary begins with some formal matters like a listing of word-forms, and a guide to pronunciation. Then, paragraph by paragraph, the meanings and uses of the word are explained. At the end

Words about words

Introduction

In the final chapter, we look at the way in which people explain the meaning of words, especially in dictionaries. Although lexicography is a practical skill, a dictionary is a systematic description of a language. In turn, it must be assumed that any such description rests on the foundations of a theoretical position, whether articulated or not.

The argument in this chapter makes something of a contrast with that in earlier work (Sinclair 1984), where I make a case against the attempt to devise a theory of lexicography. At that time, lexicography seemed to me to be almost entirely a matter of managing a number of routine factors like resources and project aims. The relevant theory was linguistic theory, pure and simple. Expertise in computation, printing, book design, reference, and other skills was required from time to time, but this was not felt to be of a theoretical nature, even when, as in computational science, theory was readily available. Lexicography was held up by the practitioners to be a largely practical matter, and theories got in the way.

However, in the later stages of compiling the Cobuild dictionary (Sinclair *et al.* 1987) it was decided to develop a new style of presenting lexicographical information. The process began in a straightforward attempt to explain the meaning and use of words in ordinary English sentences, and it ended in a radical critique of conventional lexicography. This exercise now appears to be the first step in articulating a theory of language reflexivity—the capacity of language to talk about itself. The importance of this capacity has not been properly recognized as yet, or even the extent of its occurrence in everyday usage. This chapter hopes to contribute to a better understanding of language about language.

The rationale is set out in Hanks (1987). Each entry for a word in the Cobuild dictionary begins with some formal matters like a listing of word-forms, and a guide to pronunciation. Then, paragraph by paragraph, the meanings and uses of the word are explained. At the end

of each explanation there is usually an example. To the side of the main text is an extra column with abbreviated notes on grammar and semantics.

In this chapter, I shall concentrate on the structure of explanations. The explanations lead to hypotheses about inference, metalanguage, and the general nature of lexical statement.

Here are some representative lexical statements from the Cobuild dictionary:

A **house** is a building in which people live ...

If you **defeat** someone, you win a victory over them in a contest such as a battle, game or argument.

A **pure** substance is not mixed with anything else.

If something happens **often**, it happens many times or much of the time.

Structure

These statements are first of all divisible into two principal parts:

A **house** is a building in which people live ...
If you **defeat** someone you win a victory over them in a contest ...

A **pure** substance is not mixed with anything else.
If something happens **often** it happens many times or much of the time.

The first parts of each sentence break down further into two sub-parts, shown by the type-face changes. One or more words are in bold type, and the rest is in roman. The word or phrase in bold type is called the *topic* of the sentence, and the rest of the first part can be called the *co-text*. For example, in the second statement above, *defeat* is the topic, and *if you* and *someone* constitute the co-text.

The second part of each sentence is an explanatory comment on the topic, and is called the *comment*. Comments are sometimes divisible according to the surface syntax. This is called *chunking*; in this kind of sentence, successive chunks express gradually increasing depth of detail.

There is another element of structure in each of the statements, and it can occur physically in either the first part or the second part. This element is an indication of the actual sentence structure, and is called the *operator*. In the statements above, the operators are:

- the outset of the first part: *if*;
- at the outset of the second part: *is*.

Table 1 shows the analysis so far:

FIRST PART				SECOND PART		CHUNKS
OPERATOR	CO-TEXT (1)	TOPIC	CO-TEXT (2)	OPERATOR	COMMENT	
	a	house		is	a building in which people live	1 2
if	you	defeat	someone		you win a victory over them in a contest such as ...	1 2
	a	pure	substance	is	not mixed with anything else	
if	something happens	often			it happens many times or much of the time	1 2

Table 1: Primary structure of lexical statement

Variation in co-text

There are some types of the first part that are quite different from those presented so far.

About the word itself

The statement may be about the word itself, and may not use the device of putting the word as topic in an appropriate context, for example:

You use **naturally** to indicate that you think something is very obvious.

In ordinary written English, the word *naturally* in the above example would be highlighted in some way—italics or inverted commas usually. Since it is a dictionary headword and in bold face, it is not further distinguished. However, this type of sentence is a different way of tackling the job of explanation. Other examples are:

- Naturalistic describes people or things that ...
- Meanwhile means while a particular thing is happening.
- To levitate means to rise and float in the air.
- To torture someone means ...

What people mean

The statement may be about what people mean when they use a word or phrase, rather than what the word or phrase means:

- If you describe a woman as a cow ...
- If you say that something gets up your nose ...
- If you say that something is smashing, you ...
- If you refer to someone as a pipsqueak, you ...

In these cases, it is important to point out that the opinion of the speaker is crucial to correct usage. Nothing is inherently 'smashing', in the way that it might be 'smooth' or 'strong'. The co-text includes a verb such as *describe*, *say*, *refer*, *call*, and the topic is found in a subordinate clause or a similar secondary structure. This strategy is part of the 'report' category in grammar, and typically a report contains a statement inside it.

Hence, in: 'If you describe a woman as a cow ...' you are reported as saying that the woman is a cow. *That woman is a cow* is close in structure to *a house is a building*, which is a structure already analysed. The new structure including report can be represented as follows in Table 2:

First Part				
OPERATOR	CO-TEXT			TOPIC
	REPORT	'topic'	'operator'	'comment'
If	you describe	a woman	as a	cow ...

Table 2: Report structures

The other examples can be described in similar ways. The terms *topic*, *operator*, and *comment* are re-used but in lower case and inside inverted commas to make it clear that they are embedded. Note that the comment at the lower level is the topic at the text level.

The Cobuild dictionary is sparing with explanations of this type, and only uses them when it would be misleading to ignore the subjective quality of the meaning. For example, it was implied above that 'smooth' and 'strong' were inherent qualities, but on closer inspection they are seen to be quite subjective. Something is smooth only if there is general agreement about that as a description of it. Some objective qualities of the object referred to will be relevant in deciding whether or not it can be called smooth. In contrast, if you consider something or someone 'smashing', that seems to be a very personal judgement.

Structure: verb explanations

Animate subjects

I return to the normal type of entry, to consider further the structure of the co-text. The focus is on the explanation of verbs.

In nearly every entry there is reference to a person; the sort of person who will be using English. The neutral way of referring to this person is with the pronoun *you*, and in this sense it is used many times on each page of the Cobuild dictionary.

Occasionally, though, *you* is felt not to be appropriate. The implication of using *you* is that the sentence expresses something that anyone might reasonably and normally do, so when we are explaining things which are socially undesirable, the pronoun *you* may be replaced by *someone*, for example:

- If someone burps, they make a noise ...
- If someone totters, they walk in an unsteady way ...
- If someone flings you into prison, ...

If someone **defrauds** you, ...
 If someone **burgles** a building, ...

Notice that *you* sometimes reappears as the object of the verb, so that the co-text presents the action as something happening to a person who may be the user, rather than the other way round.

There is a lot of room here for interpretation of what is taken to be socially undesirable, and the dictionary projects an evaluative view of the world through devices of this nature.

At present the use of *someone* as both subject and object is avoided, so *seduce* begins 'If you seduce someone', rather than 'If someone seduces someone'. The third possibility, 'If someone seduces you', carries an implication of this being a reasonably probable event, and so is avoided.

Someone also replaces *you* if the sentence expresses an activity which is difficult, unusual, or outside the subject's control, for example:

If someone **slips** into a particular state or activity, they change into it ...
 When someone **drowns**, ...
 When someone **sews**, ...
 If someone **tames** a wild animal or bird, ...

Where an activity is so undesirable or unusual that it would sound absurd to suggest that ordinary people do it, the choice of a subject is abandoned altogether. Instead, the explanation takes the form of a statement about the word itself, as instanced earlier in the section on 'Variation in co-text':

To **levitate** means to rise and float in the air ...
 To **torture** someone means ...

An occasional alternative to *someone* is the word *people*, which is plural and so is more natural in the description of communal activities:

When people **ski**, ...
 If people **riot**, ...
 When people **demonstrate**, ...
 If people **agree** on something, ...

The use of words like *someone* and *people* allows an important development of the co-text which is denied to the word *you*. One of the problems of explaining usage is how to decide which of these alternatives applies to each new word.

The words *someone* and *people* can be qualified by adding an illustrative or restrictive phrase:

If someone who is very ill is **sinking**, ...
 If someone, especially a child, **sneaks** on you, ...
 If someone in authority **rules** on a particular situation or problem ...

It is a small step from that to name directly a suitable subject for the topic verb:

If the police **arrest** you, ...
 When artists **exhibit** ...

The disadvantage of this last structure is that it appears by a natural sort of implication to exclude anyone other than the named people. It is thus a risky statement to make in a dictionary because the conventions of language are so easily extended. On the other hand, it is clumsy and uneconomical to keep saying 'If someone such as a policeman arrests you ...', when the likelihood of being arrested by anyone else is very small.

We must presume, then, that in the cases of direct naming it is unlikely, but not impossible, that someone other than the named person may be an appropriate subject.

Playing is associated particularly with children, and also with pet animals. Adults do it occasionally, but their play is more often expressed in clauses with an object, for example, 'Do you play chess?'. The entry in Cobuild opens thus:

When children, animals or perhaps adults **play**,

Similarly, the entry for *sting* begins:

If an insect, animal, or plant **stings** you, ...

The creatures that cause a sting are partially identified. Similarly, the entry for *lay* includes the sense:

When a bird or female animal **lays** an egg ...

The conventions of interpretation apply to non-humans in the same way as to humans, and to mixtures of human and non-human. For example, it would not be sufficiently specific to present the verb *play* in the co-text 'when you *play* ...' or even 'when people *play* ...'.

Inanimate subjects

Inanimate objects and abstract entities are dealt with by a similar set of conventions based on the word *something*. Here are some examples:

If something **glows**, ...
 If something **ensues**, ...
 If something **goes** with something else, ...

The pronoun *something* can be qualified:

If something such as success, glory or love **evades** you, ...
 If something such as an idea or subject **fits** something, ...
 If something, for example paint, **flakes**, ...
 If something unpleasant **sets in**, ...

If there is an accent on the plural, *things* can replace *something*:

If things such as ideas, beliefs or statements **sweep** a place ...

Where the likely subject is extremely restricted, it can be named:

If a play is **taken off**, ...
 When an object **breaks**, ...
 When the sun **sets**, ...

This structure allows the use of a fairly general subject as well as specific ones:

If a powerful force **tears** something from somewhere, ...
 When jelly, glue, cement, or some other soft or liquid substance **sets**.

Mixed subjects

Quite commonly a verb has a sense where the subject can be either animate or inanimate. No suitable pronoun exists in English for this usage, so we have to resort to clumsier expressions:

If someone or something **degenerates**, ...
 If someone or something **falls**, ...
 If someone or something **captivates** you, ...

With plural forms, this becomes:

When things or people **disappoint** you, ...

Operators

In the examples given so far, it will have been noticed that they all start with an operator, either *if* or *when*. The Cobuild dictionary compilers

OPERATOR	CONJUNCTION	CO-TEXT (1)						TOPIC	CO-TEXT (2)	
		MODIFIER	HEAD	BRANCHES	QUALIFIER	EXEMPLIFICATION			OBJECT	ADJUNCT
if when			you					VERB	someone	
		a (adjective) the	people						(you) (NOUN PHRASE)	(PREP PHRASE)
			NOUN	(or NOUN/ LIST)	(who is etc.) (PREP PHRASE)					
			someone	(or something)		(especially NOUN/LIST)				
			something			(such as NOUN PHRASE/LIST)				
						(for example NOUN PHRASE/ LIST)				

Table 3: Structural options in the first part of Cobuild verb explanations

chose whichever was the more comfortable expression. The result seems to be that in relation to people and animals, *if* introduces an activity which is broadly speaking within the discretion of the individual, whereas there is rather more inevitability about *when*. With *when*, the relation between subject and verb is a little more mutually determined. So, since sewing and skiing are typically human activities, they are introduced by *when*. But rioting and burbling, while still recognizably human, are not typical, and so are introduced by *if*.

In relation to inanimate objects, a similar distinction is observed. If the action seems to be inherent in the nature of the object, for example, *to break* the operator will be *when*; *when* also with *the sun sets*. If the action is only something that might happen, then *if* is used.

Summary

The structural options that have been identified so far do not present a very formidable array, and the main lines can be set out simply as in Table 3. This shows how the first parts of the verb explanations are tackled. Verb explanations are the most structurally complex, and the other parts of speech will show fewer options. Idiomatic phrases require some special attention.

Analysis of the second part

The analysis of the second part of the explanations of words proceeds along similar lines. There are references back to elements of the first part, and then a string structure of chunks adding explanatory detail.

First chunk

The first chunk of the second part of the statements is now further described, returning to the original examples of several different parts of speech, as in Table 1. In many examples, some of the words recall words in the co-text, either by repetition or other types of cohesion. These are called the *framework* of the explanation. The remainder of the first chunk is a rephrasing of the topic, and is called the *gloss*. So in the examples given, the analysis is as follows in Table 4:

First Part	Second Part			
	OPERATOR		GLOSS	
A house	is	a	building	
If you defeat someone		you	win a victory over	them
A pure substance	is		not mixed with anything else	
If something happens often		it happens	many times	

Table 4: Analysis of the second part

Of the items in the framework, *a*, *you*, and *happens* are repetitions of words in the co-text, and *it* and *them* are pronouns which refer back as follows:

it refers back to *something*;
them refers back to *someone*.

This analysis permits us to isolate the gloss element and study its role and function. In the examples above:

- building* is a replacement of *house*, and we can assume that the two words are in a recognizable semantic relation - in this case hyponymy, with *building* the superordinate. The following chunk *in which people live* provides a restriction on building, giving a classic definition:

superordinate restriction

This can be paraphrased as 'a house is a type of building. It is different from other types of building by the fact that people live in it.'

- win a victory over* replaces *defeat*. *Defeat* is a transitive verb, with the object *someone* in the co-text. In the comment, the verb is replaced by a structure with a verb, its object, and a preposition. The object of the original verb now becomes the object of the preposition.
- not mixed with anything else* replaces *pure*. Here the explanation hinges on the negative, and the semantic relation is antonymy. The adjective is replaced by a past participle and an adjunct.

- d. *many times* replaces *often*, the adverb giving way to a nominal group.

Second chunk

The second chunks of Table 1 relate to the first chunks in ways which have already been met in the first part analysis (see Table 5).

Second part, second chunk	Analysis
in which people live	qualifier of headword building
in a contest	adjunct
such as a battle, game or argument	exemplification list of nouns
or much of the time	branch

Table 5: Analysis of second chunks

Discussion

The outline analysis of the language of lexical statement shows a slight specialization of the normal conventions of English. The only physical difference is the identification of the topic by using bold face, and the analysis hinges on this. The rules of English grammar and semantics are unaffected.

The Cobuild dictionary compilers, who made an explanatory style out of a set of guide-lines, worked without restraint and used their natural choices of language to express the meanings that they wanted to convey. Several stages of consultative editing reduced the range of different structures to what is published, and work since then has made further rationalizations.

There is no particular communicative virtue in having an obviously formulaic style of explanation. Traditional dictionaries use a set of compression techniques which require specific decoding skills, and in rejecting these in favour of ordinary English it would be counterproductive to return to a formula in disguise. Rigid rules of compilation lull compilers into a false sense of security, and may obscure important distinctions of usage. For some applications, the repetitive simplicity of formulae may give greater accessibility than more accurate and varied expressions. Equally, there is some point in cutting down variation that seems over-subtle for the kind of communication in which it appears.

A dictionary is a text in which most units of discourse are very brief, and in which the overall structure is highly repetitive. The user will

naturally be led to expect that the same kind of type-face, or the same kind of phraseology, will carry the same sort of information. The reverse of this is that differences are meaningful, and so phraseology should be standardized at least to the point where differences can be justified.

Applications

There are a number of applications of the kind of analysis presented here. First of all, it will be possible to make an exhaustive comparison of actual dictionary writing. The analysis will be a professional tool with which the wording of dictionaries can be improved. Lexicographers will be able to consider alternative expressions, knowing exactly how they differ. Problem areas, for example adjectives, can be experimented with in a systematic way. The vocabulary and syntax of lexical statement will be made explicit, so that people can understand how and why meanings are explained.

Secondly, the description will be a part of the general description of English. Since the Cobuild type of explanation relies on the natural use of words, there are no new conventions to learn. The structures and meanings set out in this chapter have been derived from the dictionary text, not prescribed in advance.

The same cannot be said for most dictionary definitions, which are very obscurely structured for anyone approaching from normal English. Because of some long-established habits (for example, that the first part must be the topic only) and a great concern for compression, they require rules as if they were written in another language. So the structural resources of English are hardly available to the compilers in this kind of lexicography, and the explanations of this type of conventional dictionary are not able to be assimilated into the general repertoire of the user.

Inferences and implications

The analysis has shown that each explanation gives rise to a number of entailments, implications, and inferences. For example, the first verb explanation is phrased as follows:

If you defeat someone, you win a victory over them in a contest such as a battle, game or argument.

Immediately, we can assume that, in these circumstances (and not necessarily always),

a battle is a contest
 a game is a contest
 an argument is a contest
 you defeat someone in a contest
 defeating someone means winning a victory over them
 you win a victory over someone in a contest

The subject of *defeat* is *you*, so we can infer that defeating is not considered an unusual or reprehensible activity. The operator is *if*, signifying that it is not an inherent activity, but one that the human race is distinctly likely to indulge in from time to time. The object of *defeat* is *someone*, indicating that defeating is done to a person, but not a specially selected one.

So the combination of the general conventions of English and the particular conventions of the Cobuild dictionary are powerful in interpreting the explanations.

Conclusion

The language we use to explain the meaning of words is an important part of our linguistic repertoire. In the style studied here, it is clearly only a slight extension of the ordinary use of English. Thus, all the flexibility of a natural language is available for implications, inferences, etc. In turn, these can be developed into a set of tools by which dictionaries can be constructed and understood. All sorts of valuable semantic and structural statements can be retrieved.

For the future, this analysis offers the possibility of harnessing one of the most powerful, but least understood, features of a natural language—the features of paraphrase. The lexical statements set up equivalences, which are found in the topic and gloss categories. See Table 6 for an analysis of our original examples:

Topic (Table 1)		Gloss (Table 4)
house	=	building
defeat	=	win a victory over
pure	=	not mixed with anything else
often	=	many times

Table 6: Paraphrase

The analysis will establish that *building* is a superordinate of *house*, and the rest are synonymic. The grammatical replacement shown, for example, in *defeat* - *win a victory over* will also be made explicit. This

activity, if carried out thoroughly and accurately, is a reasonable model for the understanding of the text.

That is to say, if a person can routinely rephrase a given sentence in his or her own words and state the difference between the two sentences in a third sentence, that person will be seen as understanding the language. If a machine can perform in a similar way, the machine can reasonably be described as understanding the language. And once a machine can be seen as understanding a language, the map of information technology will have to be re-drawn.

Summing up

This book is an attempt to show that there is a lot more to learn about the English language than it was possible to imagine a few years ago. It has been suspected, of course, in all the work on idiom usage that has accumulated in language teaching materials. While grammars and dictionaries continue to report the structure of language as if it could be neatly divided, many of those people who are professionally engaged in handling language have known in their bones that the division into grammar and vocabulary obscures a very central area of meaningful organization. In fact, it may well be argued on the basis of the work in this book that when we have thoroughly pursued the patterns of co-occurrence of linguistic choices there will be little or no need for a separate residual grammar or lexicon.

That remains to be seen. Certainly, the first application of computers to the study of language corpora has uncovered a lot of new facts which have to be built in to our descriptions of languages. And it should be stressed that this book reports only the first dipping of an inquisitive toe into the vast pool of language texts. The corpus of the 1980s, although boasting a central size of 20 million words, will be seen in another decade as a relatively modest repository of evidence; the software tools increase in sophistication month by month, and must still be regarded as primitive compared with what the real needs are. Most limiting of all, our concepts, our ideas of what to expect and how to understand what we are observing, are not keeping pace with the evidence available. There is as yet little or no discussion at an international level and, beyond the Cobuild project, no thorough exploitation of corpus linguistics.